ASSIGNMENT-01

EE1204

ENGINEERING ELECTROMAGNETICS

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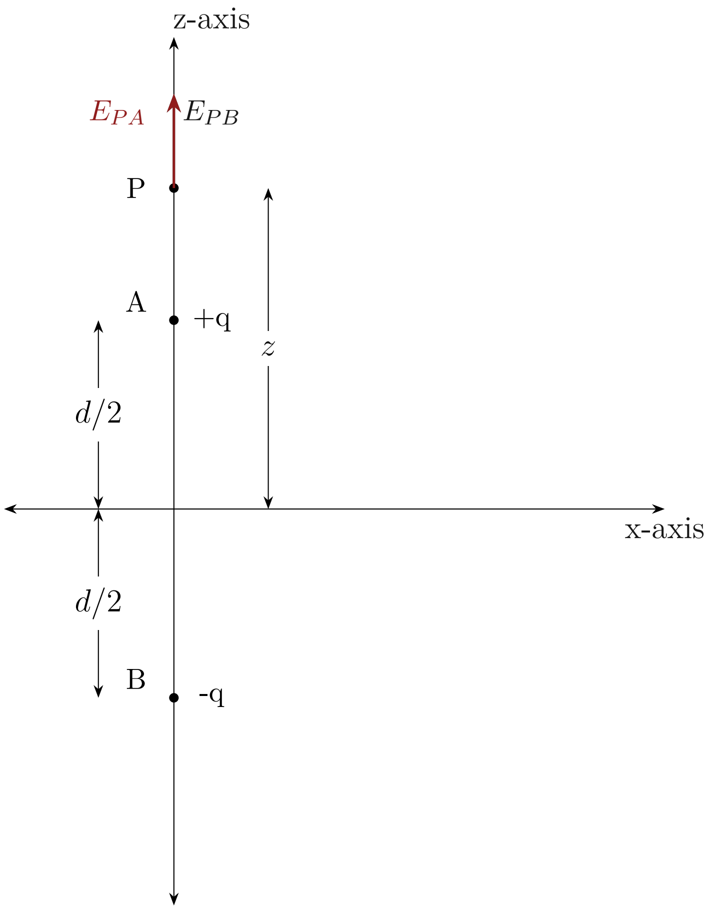
1 Two points charges of equal magnitude q are positioned at $z = \pm\frac{d}{2}$. Determine:

1.1 The electric field everywhere on the z-axis.

Solution : Below is the diagram for the question:



The position vectors for the points A,B,P are :

$$\vec{r_A} = \frac{d}{2}\hat{k} \tag{1}$$

$$\vec{r_B} = -\frac{d}{2}\hat{k} \tag{2}$$

$$\vec{r_P} = x\hat{k} \tag{3}$$

The electric field at the point P due to a charge q at point A is :

$$E_{PA} = \frac{kq}{(|r_P - r_A|)^3}\ (\vec{r_P} - \vec{r_A}) \tag{4}$$

$$E_{PA} = \frac{kq}{(x - \frac{d}{2})^3}\ (x - \frac{d}{2})\hat{k} \tag{5}$$

$$E_{PA} = \frac{kq}{(x - \frac{d}{2})^2}\ \hat{k} \tag{6}$$

where $k = \frac{1}{4\pi\epsilon_0}$
Similiarly , the electric field at the point P due to a charge q at the point B is :

$$E_{PB} = \frac{kq}{(|r_P - r_B|)^3}\ (\vec{r_P} - \vec{r_B}) \tag{7}$$

$$E_{PB} = \frac{kq}{(x + \frac{d}{2})^3}\ (x + \frac{d}{2})\hat{k} \tag{8}$$

$$E_{PB} = \frac{kq}{(x + \frac{d}{2})^2}\ \hat{k} \tag{9}$$

Now, by Principle of superposition,

$$E_P = E_{PA} + E_{PB} \tag{10}$$

$$= \frac{kq}{(x - \frac{d}{2})^2}\ \hat{k} + \frac{kq}{(x + \frac{d}{2})^2}\ \hat{k} \tag{11}$$

$$= \boxed{\frac{kq(2x^2 + \frac{d^2}{2})}{(x^2 - \frac{d^2}{4})^2}\ \hat{k}} \tag{12}$$

1.2 The electric field everywhere on the x-axis.

Solution : Below is the diagram for the question :

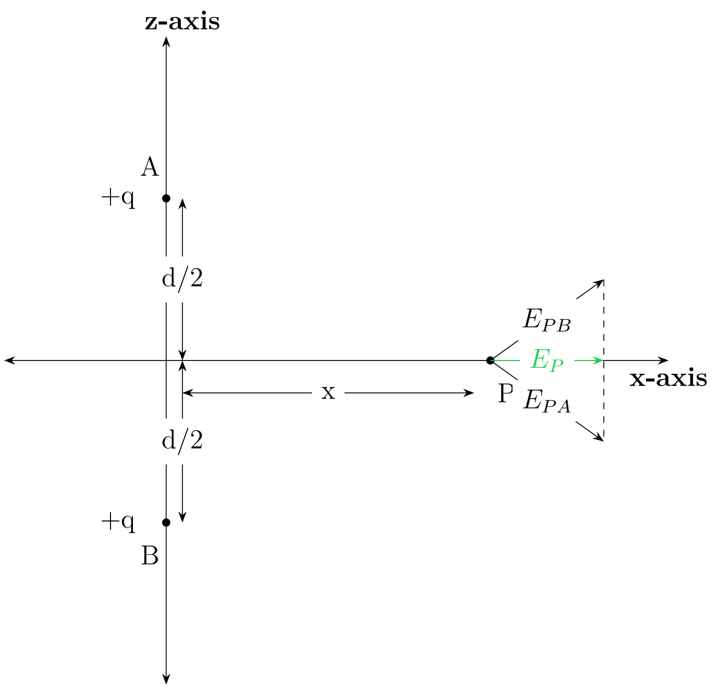


Figure 1: Description of the circuit.

The position vectors for the points A, B, P are:

$$\vec{r}_A = \frac{d}{2}\hat{k} \tag{13}$$

$$\vec{r}_B = -\frac{d}{2}\hat{k} \tag{14}$$

$$\vec{r}_P = x\hat{i} \tag{15}$$

The electric field at the point P due to a charge q at point A is:

$$\vec{E}_{PA} = \frac{kq}{|\vec{r}_P - \vec{r}_A|^3}(\vec{r}_P - \vec{r}_A) \tag{16}$$

$$= \frac{kq}{(x^2 + \frac{d^2}{4})^{3/2}}\left(x\hat{i} - \frac{d}{2}\hat{k}\right) \tag{17}$$

Similarly, the electric field at the point P due to a charge q at point B is:

$$\vec{E}_{PB} = \frac{kq}{|\vec{r}_P - \vec{r}_B|^3}(\vec{r}_P - \vec{r}_B) \tag{18}$$

$$= \frac{kq}{(x^2 + \frac{d^2}{4})^{3/2}}\left(x\hat{i} + \frac{d}{2}\hat{k}\right) \tag{19}$$

Now, by the principle of superposition:

$$\vec{E}_P = \vec{E}_{PA} + \vec{E}_{PB} \tag{20}$$

$$= \frac{kq}{(x^2 + \frac{d^2}{4})^{3/2}}\left(x\hat{i} - \frac{d}{2}\hat{k}\right) + \frac{kq}{(x^2 + \frac{d^2}{4})^{3/2}}\left(x\hat{i} + \frac{d}{2}\hat{k}\right) \tag{21}$$

$$= \frac{kq}{(x^2 + \frac{d^2}{4})^{3/2}}\left(2x\hat{i}\right) \tag{22}$$

$$= \boxed{\frac{2kqx}{(x^2 + \frac{d^2}{4})^{3/2}}\hat{i}} \tag{23}$$

1.3 Calculate the above for the above parts if the charge at $z = -\frac{d}{2}$ is $-q$ instead of q.

Solution :

For part (a) :

The electric field at the point P_z due to a charge q at point A is:

$$E_{PA} = \frac{kq}{(|r_P - r_A|)^3}(\vec{r_P} - \vec{r_A}) \tag{24}$$

$$= \frac{kq}{(x - \frac{d}{2})^3}(x - \frac{d}{2})\hat{k} \tag{25}$$

$$= \frac{kq}{(x - \frac{d}{2})^2}\hat{k} \tag{26}$$

where $k = \frac{1}{4\pi\epsilon_0}$.

Similarly, the electric field at the point P_z due to a charge $-q$ at point B is:

$$E_{PB} = \frac{k(-q)}{(|r_P - r_B|)^3}(\vec{r_P} - \vec{r_B}) \tag{27}$$

$$= -\frac{kq}{(x + \frac{d}{2})^3}(x + \frac{d}{2})\hat{k} \tag{28}$$

$$= -\frac{kq}{(x + \frac{d}{2})^2}\hat{k} \tag{29}$$

Now, by the principle of superposition:

$$
\begin{aligned}
E_P &= E_{PA} + E_{PB} && (30)\\
&= \frac{kq}{(x-\frac{d}{2})^2}\hat{k} - \frac{kq}{(x+\frac{d}{2})^2}\hat{k} && (31)\\
&= \left(\frac{kq}{(x-\frac{d}{2})^2} - \frac{kq}{(x+\frac{d}{2})^2}\right)\hat{k} && (32)\\
&= \frac{kq\left((x+\frac{d}{2})^2 - (x-\frac{d}{2})^2\right)}{(x-\frac{d}{2})^2(x+\frac{d}{2})^2}\hat{k} && (33)\\
&= \boxed{\frac{2kqdx}{(x^2-\frac{d^2}{4})^2}\hat{k}} && (34)
\end{aligned}
$$

The electric field at the point P_x due to a charge q at point A is:

$$
\begin{aligned}
\vec{E}_{PA} &= \frac{kq}{|\vec{r}_P - \vec{r}_A|^3}(\vec{r}_P - \vec{r}_A) && (35)\\
&= \frac{kq}{(x^2+\frac{d^2}{4})^{3/2}}\left(x\hat{i} - \frac{d}{2}\hat{k}\right) && (36)
\end{aligned}
$$

Similarly, the electric field at the point P_x due to a charge $-q$ at point B is:

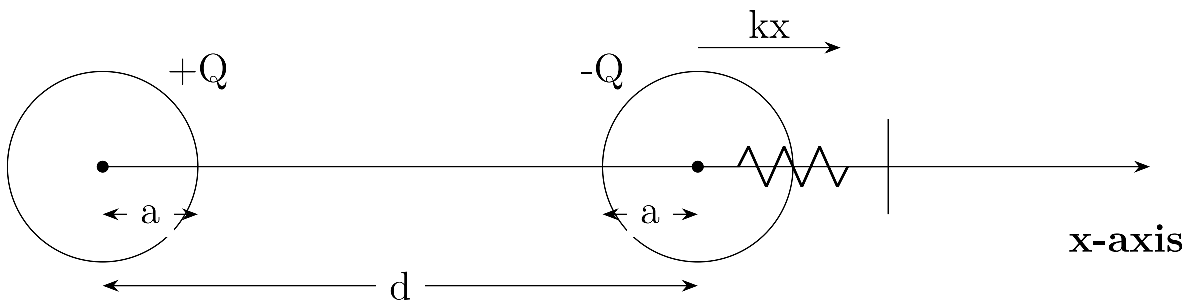
$$
\begin{aligned}
\vec{E}_{PB} &= -\frac{kq}{|\vec{r}_P - \vec{r}_B|^3}(\vec{r}_P - \vec{r}_B) && (37)\\
&= -\frac{kq}{(x^2+\frac{d^2}{4})^{3/2}}\left(x\hat{i} + \frac{d}{2}\hat{k}\right) && (38)
\end{aligned}
$$

Now, by the principle of superposition:

$$
\begin{aligned}
\vec{E}_P &= \vec{E}_{PA} + \vec{E}_{PB} && (39)\\
&= \frac{kq}{(x^2+\frac{d^2}{4})^{3/2}}\left(x\hat{i} - \frac{d}{2}\hat{k}\right) - \frac{kq}{(x^2+\frac{d^2}{4})^{3/2}}\left(x\hat{i} + \frac{d}{2}\hat{k}\right) && (40)\\
&= \boxed{-\frac{kqd}{(x^2+\frac{d^2}{4})^{3/2}}\hat{k}} && (41)
\end{aligned}
$$

2 A crude device for measuring charge consists of two small insulating spheres of radius a, one of which is fixed in position. The other is movable along the x-axis and is subject to a restraining force kx, where k is a spring constant. The uncharged spheres are centered at $x = 0$ and $x = d$, the latter fixed. If the spheres are given equal and opposite charges of $\frac{Q}{C}$, obtain the expression by which Q may be found as a function of x.Determine the maximum charge that can be measured in terms of ϵ_0, k, and d, and state the separation of the spheres. What happens if a larger charge is applied?

Solution : The diagram for the question is below : To analyze the system, we denote



the positions of the two spheres as vectors: - The position of the fixed sphere (centered at $x = 0$) is given by:

$$\vec{r}_A = 0\hat{i}$$

- The position of the movable sphere (centered at $x = d$) is given by:

$$\vec{r}_B = d\hat{i}$$

The distance between the centers of the spheres when the movable sphere is at position x is:

$$\vec{r}_{AB} = \vec{r}_B - \vec{r}_A = (d - x)\hat{i}$$

The magnitude of the distance is:

$$|\vec{r}_{AB}| = d - x$$

The electrostatic force $\vec{F}_e$ acting on the movable sphere due to the charge Q at the fixed sphere is given by Coulomb's law:

$$\vec{F}_e = \frac{1}{4\pi\epsilon_0}\frac{Q^2}{|\vec{r}_{AB}|^2}\hat{r}_{AB}$$

where $\hat{r}_{AB}$ is the unit vector in the direction of $\vec{r}_{AB}$:

$$\hat{r}_{AB} = \frac{\vec{r}_{AB}}{|\vec{r}_{AB}|} = \hat{i}$$

Thus, the electrostatic force becomes:

$$\vec{F}_e = \frac{1}{4\pi\epsilon_0}\frac{Q^2}{(d-x)^2}\hat{i}$$

The restoring force $\vec{F}_s$ from the spring is:

$$\vec{F}_s = -kx\hat{i}$$

At equilibrium, the total force acting on the movable sphere must equal zero:

$$\vec{F}_{total} = \vec{F}_e + \vec{F}_s = 0$$

Substituting the forces gives:

$$\frac{1}{4\pi\epsilon_0}\frac{Q^2}{(d-x)^2}\hat{i} - kx\hat{i} = 0$$

Removing the unit vector $\hat{i}$ from the equation results in:

$$\frac{1}{4\pi\epsilon_0}\frac{Q^2}{(d-x)^2} - kx = 0$$

Rearranging this equation yields:

$$\frac{Q^2}{(d-x)^2} = 4\pi\epsilon_0 kx$$

$$Q^2 = 4\pi\epsilon_0 kx(d-x)^2$$

Therefore, the expression for Q as a function of x is:

$$Q = \sqrt{4\pi\epsilon_0 kx(d-x)^2}$$

To find the maximum charge, we differentiate Q^2 with respect to x:

$$Q^2 = 4\pi\epsilon_0 kx(d-x)^2$$

Let $f(x) = Q^2 = 4\pi\epsilon_0 kx(d-x)^2$. To find the maximum, we calculate the derivative:

$$\begin{aligned}\frac{df}{dx} &= 4\pi\epsilon_0 k\left((d-x)^2 + x\cdot 2(d-x)(-1)\right)\\ &= 4\pi\epsilon_0 k\left((d-x)^2 - 2x(d-x)\right)\\ &= 4\pi\epsilon_0 k\left(d^2 - 2dx + x^2 - 2dx + 2x^2\right)\\ &= 4\pi\epsilon_0 k\left(d^2 - 4dx + 3x^2\right)\end{aligned}$$

Setting the derivative to zero for maximization:

$$d^2 - 4dx + 3x^2 = 0$$

Using the quadratic formula, we have:

$$\begin{aligned} x &= \frac{4d \pm \sqrt{(4d)^2 - 4 \cdot 3 \cdot d^2}}{2 \cdot 3} \\ &= \frac{4d \pm \sqrt{16d^2 - 12d^2}}{6} \\ &= \frac{4d \pm 2d}{6} \end{aligned}$$

This gives two solutions:

$$x = d \quad \text{and} \quad x = \frac{d}{3}$$

The relevant solution for maximum charge is $x = \frac{d}{3}$.
Substituting $x = \frac{d}{3}$ back into the expression for Q:

$$\begin{aligned} Q_{max} &= \sqrt{4\pi\epsilon_0 k \cdot \frac{d}{3}\left(d - \frac{d}{3}\right)^2} \\ &= \sqrt{4\pi\epsilon_0 k \cdot \frac{d}{3} \cdot \left(\frac{2d}{3}\right)^2} \\ &= \sqrt{4\pi\epsilon_0 k \cdot \frac{d}{3} \cdot \frac{4d^2}{9}} \\ &= \boxed{\sqrt{\frac{16\pi\epsilon_0 k d^3}{27}}} \end{aligned}$$

3 A flux density field is given as

$$\mathbf{F_1} = 5\hat{a}_z.$$

The task is to evaluate the outward flux of $\mathbf{F_1}$ through the hemispherical surface defined by $r = a$,

$$0 < \theta < \frac{\pi}{2}, \quad 0 < \phi < 2\pi.$$

Next, consider what simple observation would have saved a lot of work in the previous part. Identifying symmetries or using alternative methods can simplify the calculations significantly. Now suppose the field is given by

$$\mathbf{F_2} = 5z\hat{a}_z.$$

Using the appropriate surface integrals, determine the net outward flux of $\mathbf{F_2}$ through the closed surface consisting of the hemisphere from the first part and its circular base in the xy-plane. Finally, repeat the previous calculation by applying the divergence theorem and evaluating an appropriate volume integral. This approach should confirm the result obtained through direct surface integration.

Solution :

3.1 Flux of $F_1 = 5\hat{a}_z$ through the hemispherical surface

The outward flux Φ through a surface S is given by:

$$\Phi = \iint_S \mathbf{F} \cdot d\mathbf{S}$$

For the hemispherical surface defined by $r = a$, $0 < \theta < \frac{\pi}{2}$, and $0 < \phi < 2\pi$:

1. **Surface Area Element**: In spherical coordinates, the differential area element on the hemispherical surface is:

$$d\mathbf{S} = \hat{r}\, r^2 \sin(\theta)\, d\theta\, d\phi$$

Here, $\hat{r}$ is the outward normal vector. For the hemisphere, $\hat{r} = \sin(\theta)\cos(\phi)\hat{x} + \sin(\theta)\sin(\phi)\hat{y} + \cos(\theta)\hat{z}$.

2. **Evaluate the Dot Product**:

$$\mathbf{F_1} = 5\hat{z} = 5\cos(\theta)\hat{r} + 0\hat{x} + 0\hat{y}$$

Calculating the dot product:

$$\mathbf{F_1} \cdot d\mathbf{S} = 5\hat{z} \cdot \hat{r}\, r^2 \sin(\theta)\, d\theta\, d\phi = 5\cos(\theta) r^2 \sin(\theta)\, d\theta\, d\phi$$

Substituting $r = a$:

$$\Phi = \iint_S 5a^2 \cos(\theta)\sin(\theta)\, d\theta\, d\phi$$

3. **Limits of Integration**:

- θ from 0 to $\frac{\pi}{2}$
- ϕ from 0 to 2π

4. **Integrating**:

$$\Phi = 5a^2 \int_0^{2\pi} d\phi \int_0^{\frac{\pi}{2}} \cos(\theta)\sin(\theta)\, d\theta$$

The integral $\int_0^{\frac{\pi}{2}} \cos(\theta)\sin(\theta)\, d\theta = \frac{1}{2}$.

So,

$$\Phi = 5a^2 \cdot 2\pi \cdot \frac{1}{2} = 5\pi a^2$$

3.2 Observation to Simplify

A simple observation that could have saved work: the vector field F_1 is uniform and directed along the z-axis, leading to symmetry. The flux through the curved surface can often be simplified by recognizing the symmetry of the field and the surface.

3.3 Flux of $F_2 = 5z\hat{a}_z$

For F_2, the flux through the closed surface (hemisphere + circular base) is given by:

$$\Phi = \iint_S F_2 \cdot dS$$

1. **Curved Hemisphere**:

   The calculations for the hemispherical surface remain similar to part 1, with $F_2 = 5z\hat{a}_z$.

2. **Base in the xy-plane**:

   On the circular base, where $z = 0$:

   $$F_2 = 0\hat{a}_z$$

   Thus, the contribution to the flux from the base is zero.

   Therefore, the total outward flux through the closed surface is the same as through the hemisphere:

   $$\Phi = 5\pi a^2$$

3.4 Divergence Theorem

Using the divergence theorem:

$$\Phi = \iiint_V (\nabla \cdot F_2)\, dV$$

Calculate the divergence:

$$\nabla \cdot F_2 = \frac{\partial(5z)}{\partial z} = 5$$

The volume V of the hemisphere can be calculated as:

$$V = \frac{1}{2} \cdot \frac{4}{3}\pi a^3 = \frac{2}{3}\pi a^3$$

Thus,

$$\Phi = \iiint_V 5\, dV = 5 \cdot \frac{2}{3}\pi a^3 = \frac{10}{3}\pi a^3$$

4 An infinitely long cylindrical dielectric of radius b contains charge within its volume with a charge density given by:

$$\rho_v = a\rho^2$$

where a is a constant. The goal is to determine the electric field strength E both inside and outside the cylinder.

Solution :

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

where ϵ_0 is the permittivity of the dielectric, and Q_{enc} is the charge enclosed by the Gaussian surface.

4.1 Field Inside the Cylinder $(0 \leq \rho < b)$

We choose a Gaussian cylindrical surface of radius ρ and length L, coaxial with the charged cylinder.

4.1.1 1. Enclosed Charge

The charge enclosed within the Gaussian surface is:

$$Q_{\text{enc}} = \int_{\text{volume}} \rho_v dV$$

Since the charge density is given as:

$$\rho_v = a\rho^2$$

we write the volume element in cylindrical coordinates:

$$dV = (2\pi\rho' d\rho' L)$$

Thus, the enclosed charge is:

$$Q_{\text{enc}} = \int_0^\rho a\rho'^2 (2\pi\rho' L) d\rho'$$

$$= 2\pi aL \int_0^\rho \rho'^3 d\rho'$$

$$= 2\pi aL \left[\frac{\rho'^4}{4}\right]_0^\rho$$

$$= \frac{2\pi aL}{4}\rho^4 = \frac{\pi aL}{2}\rho^4$$

4.1.2 Apply Gauss's Law

The left-hand side of Gauss's Law is:

$$\oint \mathbf{E} \cdot d\mathbf{A} = E(2\pi\rho L)$$

Equating to $Q_{\text{enc}}/\epsilon_0$:

$$E(2\pi\rho L) = \frac{\pi aL}{2\epsilon_0}\rho^4$$

Solving for E:

$$E = \frac{a}{4\epsilon_0}\rho^3$$

So, inside the cylinder:

$$\mathbf{E} = \frac{a}{4\epsilon_0}\rho^3\hat{\rho}$$

4.2 Field Outside the Cylinder ($\rho \geq b$)

For $\rho \geq b$, the total enclosed charge is the charge within the entire cylinder ($0 \leq \rho' \leq b$):

$$Q_{\text{enc}} = \frac{\pi aL}{2}b^4$$

Using Gauss's Law for a Gaussian surface of radius ρ:

$$E(2\pi\rho L) = \frac{\pi aL}{2\epsilon_0}b^4$$

Solving for E:

$$E = \frac{a}{4\epsilon_0}\frac{b^4}{\rho}$$

So, outside the cylinder:

$$\mathbf{E} = \frac{a}{4\epsilon_0}\frac{b^4}{\rho}\hat{\rho}$$

$$\boxed{\mathbf{E} = \begin{cases} \frac{a}{4\epsilon_0}\rho^3\hat{\rho}, & 0 \leq \rho < b \\ \frac{a}{4\epsilon_0}\frac{b^4}{\rho}\hat{\rho}, & \rho \geq b \end{cases}}$$

5 Given the vector field in cylindrical coordinates:

$$\mathbf{F} = \left(\frac{40}{s^2+1} + 3(\cos\phi + \sin\phi) \right) \hat{s} + 3(\cos\phi - \sin\phi)\hat{\phi} - 2\hat{z}$$

5.1 Compute and plot the magnitude $|\mathbf{F}|$ as a function of ϕ for $s = 3$.

Solution : The magnitude of the vector field is given by:

$$|\mathbf{F}| = \sqrt{F_s^2 + F_\phi^2 + F_z^2}$$

where:

$$F_s = \frac{40}{s^2+1} + 3(\cos\phi + \sin\phi)$$

$$F_\phi = 3(\cos\phi - \sin\phi)$$

$$F_z = -2$$

For $s = 3$:

$$F_s = \frac{40}{3^2+1} + 3(\cos\phi + \sin\phi) = \frac{40}{10} + 3(\cos\phi + \sin\phi) = 4 + 3(\cos\phi + \sin\phi)$$

Thus, the magnitude of $\mathbf{F}$ simplifies to:

$$|\mathbf{F}| = \sqrt{(4 + 3(\cos\phi + \sin\phi))^2 + (3(\cos\phi - \sin\phi))^2 + (-2)^2}$$

$$|\mathbf{F}| = \sqrt{(3(\cos\phi + \sin\phi)^2 + 24(\cos\phi + \sin\phi) + 16 + (3(\cos\phi - \sin\phi))^2 + 4}$$

$$|\mathbf{F}| = \sqrt{18 + 16 + 4 + 24(\cos\phi + \sin\phi)}$$

$$\boxed{|\mathbf{F}| = \sqrt{38 + 24(\cos\phi + \sin\phi)}}$$

5.1.1 The Plot

Below is the plot for $\mathbf{F}$:

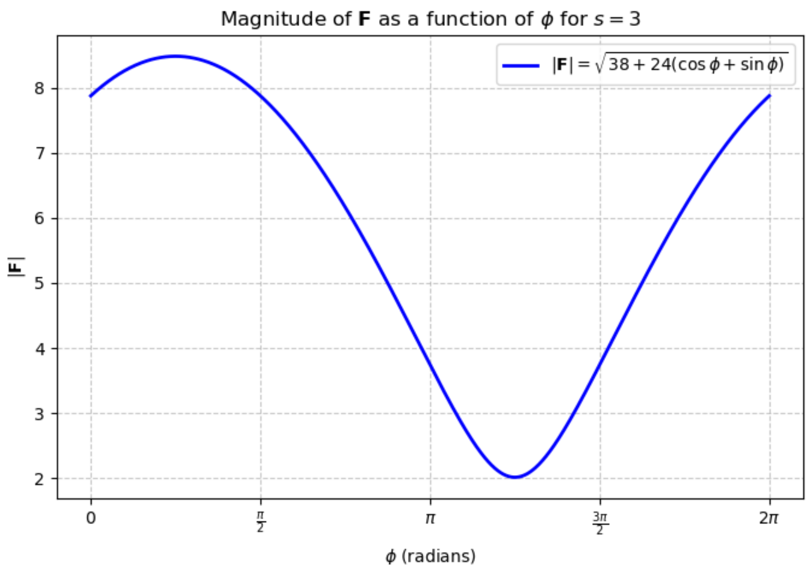


Figure 2: Plot of $|\mathbf{F}|$ as a function of ϕ for $s = 3$

5.1.2 Python Code for Plotting

The following code generates the plot of $|\mathbf{F}|$ as a function of ϕ:

```
import numpy as np  # Import NumPy for mathematical functions
import matplotlib.pyplot as plt  # Import Matplotlib for plotting

# Define the function to compute the magnitude of F
def F_magnitude(phi):
    """
    Computes the magnitude of the vector field |F| as a function of phi.

    Parameters:
        phi (array): Array of phi values in radians.

    Returns:
        array: Computed magnitude values.
    """
    return np.sqrt(38 + 24 * (np.cos(phi) + np.sin(phi)))

# Generate an array of phi values from 0 to 2*pi (one full cycle) with 1000 points
phi_values = np.linspace(0, 2 * np.pi, 1000)  # 1000 points for a very smooth curve

# Compute the magnitude for each phi value
F_values = F_magnitude(phi_values)

# Create a figure and plot the function
plt.figure(figsize=(8, 5))  # Set the figure size
plt.plot(phi_values, F_values, label=r'|F| = sqrt{38 + 24(cos(phi) + sin(phi))}'
,color='b', linewidth=2)

# Mark important points at 0, /2, , 3 /2, and 2
phi_ticks = [0, np.pi/2, np.pi, 3*np.pi/2, 2*np.pi]
phi_labels = [r'0', r'\pi/2', r'pi', r'3pi/2', r'2 pi']
plt.xticks(phi_ticks, phi_labels)

```

```python
# Add labels and title to the plot
plt.xlabel(r'phi (radians)')  # X-axis label (phi in radians)
plt.ylabel(r'|F|')  # Y-axis label (magnitude of F)
plt.title(r'Magnitude of F as a function of phi for s=3')  # Title of the plot

# Add legend and grid for better visualization
plt.legend(loc='upper right', fontsize=10, frameon=True)
plt.grid(True, linestyle='--', alpha=0.7)

# Show the plot
plt.show()
```

5.2 Compute and plot the magnitude $|\mathbf{F}|$ as a function of s for $\phi = 45°$.

Solution : For $\phi = 45°$ ($\phi = \frac{\pi}{4}$):

$$\cos\frac{\pi}{4} = \sin\frac{\pi}{4} = \frac{\sqrt{2}}{2}$$

$$F_s = \frac{40}{s^2+1} + 3\sqrt{2}, \quad F_\phi = 0, \quad F_z = -2$$

$$|\mathbf{F}| = \sqrt{\left(\frac{40}{s^2+1} + 3\sqrt{2}\right)^2 + (-2)^2}$$

$$\boxed{|\mathbf{F}| = \sqrt{\left(\frac{40}{s^2+1} + 3\sqrt{2}\right)^2 + 4}}$$

5.2.1 The Plot

Below is the plot for $\mathbf{F}$:

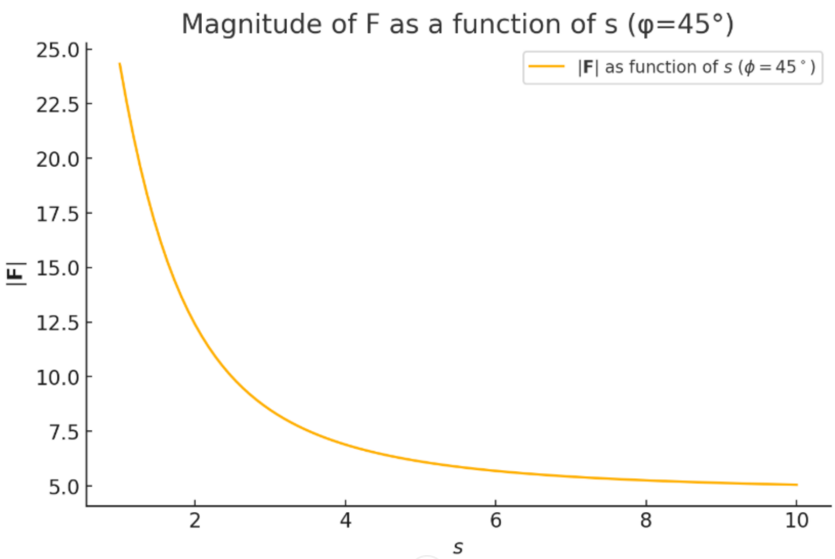


Figure 3: Plot of $|\mathbf{F}|$ as a function of s for $\phi = 45°$

5.2.2 Plot of $|\mathbf{F}|$ as a function of s

```
# Define s values
s = np.linspace(1, 10, 100)
phi_fixed = np.pi / 4  # 45 degrees

# Compute F_s, F_phi, and F_z
F_s_b = 40 / (s**2 + 1) + 3 * np.sqrt(2)
F_phi_b = 0  # Since cos(45) - sin(45) = 0
F_z_b = -2

# Compute magnitude |F|
F_mag_b = np.sqrt(F_s_b**2 + F_phi_b**2 + F_z_b**2)

# Plot
plt.figure(figsize=(8, 5))
plt.plot(s, F_mag_b, label=r'$|\mathbf{F}|$ as function of $s$ ($\phi=45^\circ$)')
plt.xlabel(r'$s$')
plt.ylabel(r'$|\mathbf{F}|$')
plt.title('Magnitude of F as a function of s ( =45°)')
plt.legend()
plt.grid()
plt.show()
```

5.3 Calculate the divergence of F.

In cylindrical coordinates, the divergence of a vector field $\mathbf{F} = F_s\hat{s} + F_\phi\hat{\phi} + F_z\hat{z}$ is given by:

$$\nabla \cdot \mathbf{F} = \frac{1}{s}\frac{\partial}{\partial s}(sF_s) + \frac{1}{s}\frac{\partial F_\phi}{\partial \phi} + \frac{\partial F_z}{\partial z}$$

$$F_s = \frac{40}{s^2+1} + 3(\cos\phi + \sin\phi), \quad F_\phi = 3(\cos\phi - \sin\phi), \quad F_z = -2$$

$$sF_s = s\left(\frac{40}{s^2+1} + 3(\cos\phi + \sin\phi)\right)$$

Applying the quotient rule:

$$\frac{\partial}{\partial s}\left(s\frac{40}{s^2+1}\right) = \frac{40(s^2+1) - 40s(2s)}{(s^2+1)^2} = \frac{40 - 40s^2}{(s^2+1)^2}$$

For the other term:

$$\frac{\partial}{\partial s}[s(3\cos\phi + 3\sin\phi)] = 3\cos\phi + 3\sin\phi$$

Thus,

$$\frac{1}{s}\frac{\partial}{\partial s}(sF_s) = \frac{1}{s}\left(\frac{40 - 40s^2}{(s^2+1)^2} + 3\cos\phi + 3\sin\phi\right)$$

$$\frac{\partial}{\partial \phi}(3\cos\phi - 3\sin\phi) = -3\sin\phi - 3\cos\phi$$

$$\frac{1}{s}\frac{\partial F_\phi}{\partial \phi} = \frac{-3\sin\phi - 3\cos\phi}{s}$$

Since $F_z = -2$ is constant,

$$\frac{\partial F_z}{\partial z} = 0$$

$$\nabla \cdot \mathbf{F} = \frac{1}{s}\left(\frac{40 - 40s^2}{(s^2+1)^2} + 3\cos\phi + 3\sin\phi\right) + \frac{-3\sin\phi - 3\cos\phi}{s}$$

$$\boxed{\nabla \cdot \mathbf{F} = \frac{40 - 40s^2}{s(s^2+1)^2}}$$

5.4 Calculate the curl of F and verify if the field is conservative .

The curl of **F** in cylindrical coordinates is:

$$\nabla \times \mathbf{F} = \begin{vmatrix} \hat{s} & \frac{1}{s}\hat{\phi} & \hat{z} \\ \frac{\partial}{\partial s} & \frac{1}{s}\frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ F_s & F_\phi & F_z \end{vmatrix}$$

The components of the curl are:

$$(\nabla \times \mathbf{F})_s = \frac{1}{s}\frac{\partial F_z}{\partial \phi} - \frac{\partial F_\phi}{\partial z}$$

$$(\nabla \times \mathbf{F})_\phi = \frac{\partial F_s}{\partial z} - \frac{\partial F_z}{\partial s}$$

$$(\nabla \times \mathbf{F})_z = \frac{1}{s}\left(\frac{\partial}{\partial s}(sF_\phi) - \frac{\partial F_s}{\partial \phi}\right)$$

$$\frac{1}{s}\frac{\partial F_z}{\partial \phi} - \frac{\partial F_\phi}{\partial z}$$

Since $F_z = -2$ is constant, $\frac{\partial F_z}{\partial \phi} = 0$. Also, F_ϕ does not depend on z, so $\frac{\partial F_\phi}{\partial z} = 0$.

$$(\nabla \times \mathbf{F})_s = 0$$

$$\frac{\partial F_s}{\partial z} - \frac{\partial F_z}{\partial s}$$

Since neither F_s nor F_z depends on z, we get:

$$(\nabla \times \mathbf{F})_\phi = 0$$

$$\frac{1}{s}\left(\frac{\partial}{\partial s}(sF_\phi) - \frac{\partial F_s}{\partial \phi}\right)$$

Computing:

$$\frac{\partial}{\partial s}(sF_\phi) = \frac{\partial}{\partial s}(s(3\cos\phi - 3\sin\phi)) = 3\cos\phi - 3\sin\phi$$

$$\frac{\partial F_s}{\partial \phi} = -3\sin\phi + 3\cos\phi$$

$$(\nabla \times \mathbf{F})_z = \frac{1}{s}(3\cos\phi - 3\sin\phi - (3\cos\phi - 3\sin\phi)) = 0$$

$$\boxed{\nabla \times \mathbf{F} = 0}$$

Since the curl is zero, the field is **conservative**, meaning there exists a scalar potential Φ such that:

$$\mathbf{F} = \nabla\Phi$$

6 Refer to the charge distribution below :

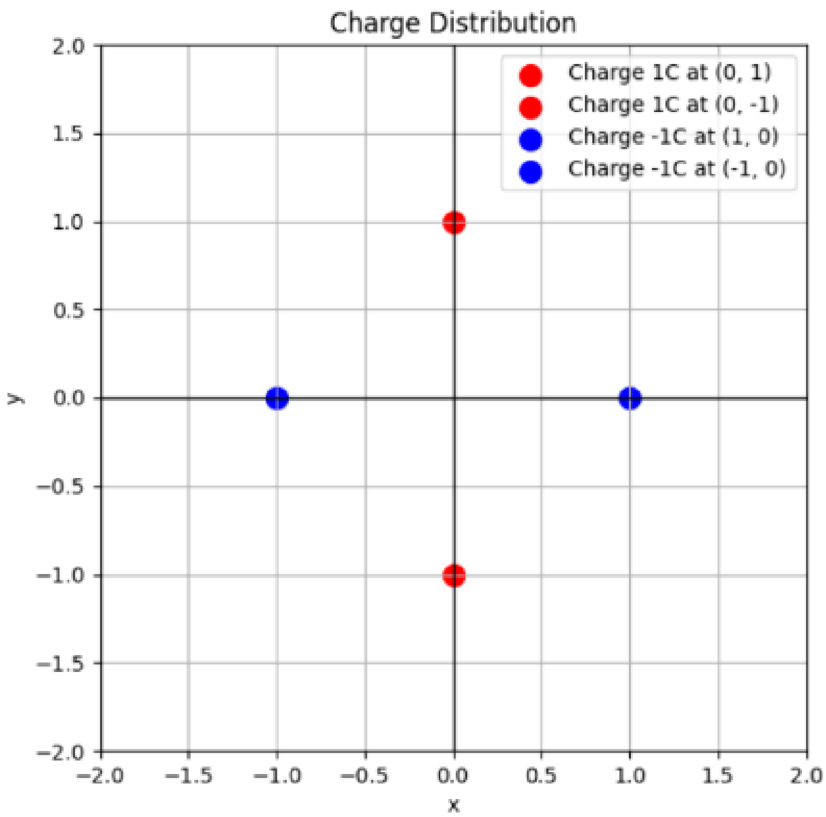


Figure 4: Question Figure

6.1 Plot the electric field .

```
import numpy as np
import matplotlib.pyplot as plt  # Import required libraries

# Constants
k = 9e9  # Coulomb's constant (N·m²/C²)

# Define charge positions and values based on the given image
charges = [
    (0, 1, 1),   # Charge of +1C at (0,1)
    (0, -1, 1),  # Charge of +1C at (0,-1)
    (1, 0, -1),  # Charge of -1C at (1,0)
    (-1, 0, -1)  # Charge of -1C at (-1,0)
]

# Define a grid of points where we will compute the electric field
x_vals = np.linspace(-2, 2, 20)  # X-coordinates from -2 to 2 with 20 points
y_vals = np.linspace(-2, 2, 20)  # Y-coordinates from -2 to 2 with 20 points
X, Y = np.meshgrid(x_vals, y_vals)  # Create a 2D grid of points

# Initialize electric field components (Ex, Ey) as zero arrays
Ex = np.zeros(X.shape)  # X-component of electric field
Ey = np.zeros(Y.shape)  # Y-component of electric field

# Loop through each charge to compute its contribution to the field
for (xq, yq, q) in charges:
    dx = X - xq  # X-distance from charge to grid points
    dy = Y - yq  # Y-distance from charge to grid points
    r = np.sqrt(dx**2 + dy**2)  # Compute the Euclidean distance from charge

    r[r < 0.2] = 0.2  # Prevent division by zero by setting a minimum radius

    # Compute the magnitude of the electric field using Coulomb's law: E = kq / r²
    E = k * q / r**2

    # Compute the electric field components Ex and Ey
    Ex += E * (dx / r)  # Contribution of this charge to the Ex field
    Ey += E * (dy / r)  # Contribution of this charge to the Ey field

# Normalize field vectors for better visualization (scale all vectors to
unit length)
magnitude = np.sqrt(Ex**2 + Ey**2)  # Compute field magnitude
Ex /= magnitude  # Normalize Ex
Ey /= magnitude  # Normalize Ey

# Plot the electric field using quiver plot (vector arrows)
plt.figure(figsize=(6, 6))  # Set figure size
plt.quiver(X, Y, Ex, Ey, color='b', scale=20)  # Plot electric field vectors

# Plot the charge locations with colors (Red for +1C, Blue for -1C)
plt.scatter([0, 0, 1, -1], [1, -1, 0, 0], c=['red', 'red', 'blue', 'blue'],
s=100, label='Charges')
```

```
# Label axes
plt.xlabel('x')
plt.ylabel('y')

# Set title for the plot
plt.title('Electric Field of Given Charge Distribution')

# Show legend
plt.legend()

# Display grid for reference
plt.grid()

# Show the plot
plt.show()
```

6.1.1 Plot

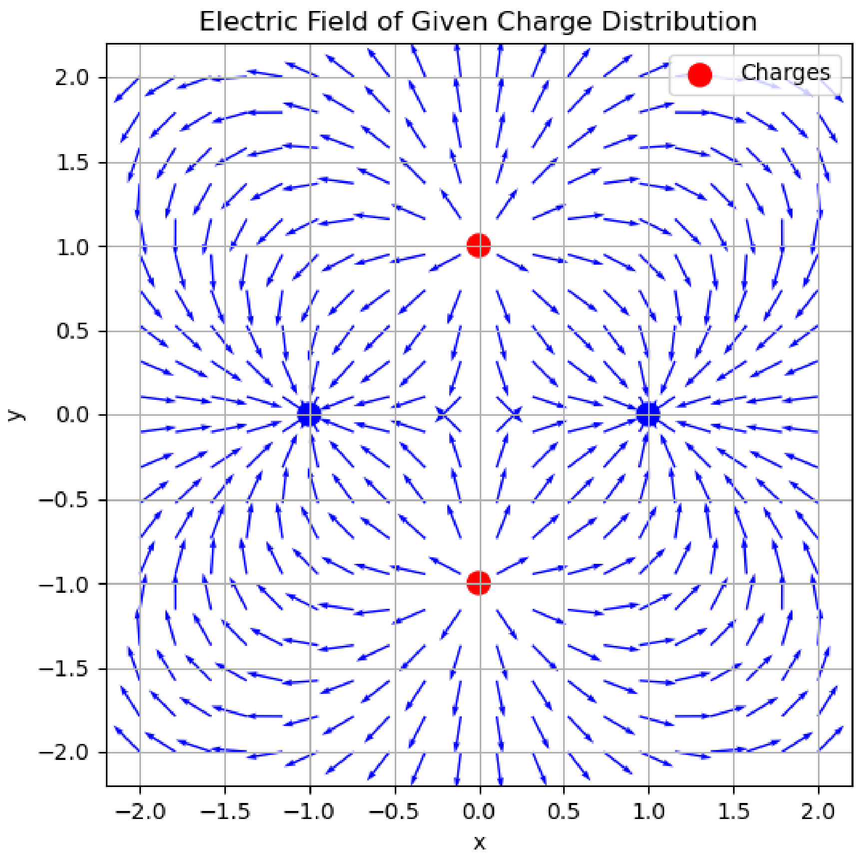


Figure 5: Electric Field

6.2 Plot the potential with magnitude represented along the z-axis.

6.2.1 Plot the potential

```
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

# Constants
k = 9e9  # Coulomb's constant (N·m²/C²)

# Define charge positions and values
charges = [
    (0, 1, 1),   # Charge of +1C at (0,1)
    (0, -1, 1),  # Charge of +1C at (0,-1)
    (1, 0, -1),  # Charge of -1C at (1,0)
    (-1, 0, -1)  # Charge of -1C at (-1,0)
]

# Define a grid of points where we will compute the potential
x_vals = np.linspace(-2, 2, 50)  # X-coordinates
y_vals = np.linspace(-2, 2, 50)  # Y-coordinates
X, Y = np.meshgrid(x_vals, y_vals)  # Create a 2D grid

# Initialize electric potential array
V = np.zeros(X.shape)

# Compute the potential at each grid point
for (xq, yq, q) in charges:
    dx = X - xq  # X-distance from charge
    dy = Y - yq  # Y-distance from charge
    r = np.sqrt(dx**2 + dy**2)  # Distance from charge

    r[r < 0.2] = 0.2  # Prevent division by zero
    V += k * q / r  # Compute potential using V = kq/r

# Create 3D plot
fig = plt.figure(figsize=(8, 6))
ax = fig.add_subplot(111, projection='3d')

# Plot the surface
ax.plot_surface(X, Y, V, cmap='coolwarm', edgecolor='none')

# Label axes
ax.set_xlabel('X')
ax.set_ylabel('Y')
ax.set_zlabel('Potential V')
ax.set_title('Electric Potential Surface')

# Show the plot
plt.show()
```

6.2.2 Plot

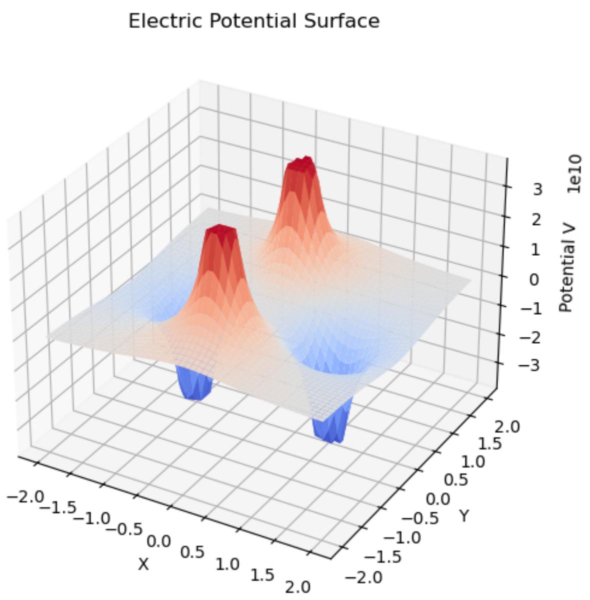


Figure 6: Electric Potential

6.3 Compute the potential energy of the configuration

The electrostatic potential energy of a system of point charges is given by:

$$U = k \sum_{i<j} \frac{q_i q_j}{r_{ij}} \tag{42}$$

where:

- U is the total electrostatic potential energy,
- $k = 9 \times 10^9$ Nm²/C² (Coulomb's constant),
- q_i, q_j are the interacting charges,
- r_{ij} is the distance between charges q_i and q_j.

We consider four point charges located at:

$$(0, 1), (0, -1), (1, 0), (-1, 0)$$

with values $+1C$ and $-1C$. The pairwise distances are:

$$r_{12} = 2, \quad r_{13} = \sqrt{2}, \quad r_{14} = \sqrt{2}, \quad r_{23} = \sqrt{2}, \quad r_{24} = \sqrt{2}, \quad r_{34} = 2$$

Substituting the values:

$$U = k\left(\frac{(1)(1)}{2} + \frac{(1)(-1)}{\sqrt{2}} + \frac{(1)(-1)}{\sqrt{2}} + \frac{(1)(-1)}{\sqrt{2}} + \frac{(1)(-1)}{\sqrt{2}} + \frac{(-1)(-1)}{2}\right)$$

$$U = (9 \times 10^9)\left(\frac{1}{2} - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} + \frac{1}{2}\right)$$

$$U = (9 \times 10^9)\left(1 - 4 \times \frac{1}{\sqrt{2}}\right)$$

Approximating $\frac{1}{\sqrt{2}} \approx 0.707$:

$$U = (9 \times 10^9) \times (1 - 2.828)$$

$$U = (9 \times 10^9) \times (-1.828)$$

$$\boxed{U \approx -1.645 \times 10^{10} \text{ Joules}}$$

6.4 Use Gauss's Law to verify the divergence of the electric field.

Gauss's Law states that the total electric flux through a closed surface is proportional to the total charge enclosed:

$$\oint_S \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{enc}}}{\varepsilon_0} \tag{43}$$

where:

- $\mathbf{E}$ is the electric field,
- $d\mathbf{A}$ is the differential area element,
- Q_{enc} is the total charge enclosed within the surface,
- ε_0 is the permittivity of free space (8.85×10^{-12} C²/N · m²).

The divergence theorem states:

$$\oint_S \mathbf{E} \cdot d\mathbf{A} = \int_V (\nabla \cdot \mathbf{E}) dV \tag{44}$$

Comparing with Gauss's Law:

$$\int_V (\nabla \cdot \mathbf{E}) dV = \frac{Q_{\text{enc}}}{\varepsilon_0} \tag{45}$$

Thus, the differential form of Gauss's Law is:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0} \tag{46}$$

where ρ is the charge density.

Given the electric field due to a point charge:

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r} \tag{47}$$

we express $\mathbf{E}$ in Cartesian coordinates:

$$E_x = \frac{1}{4\pi\varepsilon_0}\frac{qx}{(x^2+y^2+z^2)^{3/2}}, \quad E_y = \frac{1}{4\pi\varepsilon_0}\frac{qy}{(x^2+y^2+z^2)^{3/2}}, \quad E_z = \frac{1}{4\pi\varepsilon_0}\frac{qz}{(x^2+y^2+z^2)^{3/2}} \tag{48}$$

The divergence of $\mathbf{E}$ is computed as:

$$\nabla\cdot\mathbf{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} \tag{49}$$

Computing each term:

$$\frac{\partial E_x}{\partial x} = \frac{q}{4\pi\varepsilon_0}\frac{(y^2+z^2-2x^2)}{(x^2+y^2+z^2)^{5/2}} \tag{50}$$

$$\frac{\partial E_y}{\partial y} = \frac{q}{4\pi\varepsilon_0}\frac{(x^2+z^2-2y^2)}{(x^2+y^2+z^2)^{5/2}} \tag{51}$$

$$\frac{\partial E_z}{\partial z} = \frac{q}{4\pi\varepsilon_0}\frac{(x^2+y^2-2z^2)}{(x^2+y^2+z^2)^{5/2}} \tag{52}$$

Summing these terms:

$$\nabla\cdot\mathbf{E} = \frac{q}{4\pi\varepsilon_0}\frac{(y^2+z^2-2x^2)+(x^2+z^2-2y^2)+(x^2+y^2-2z^2)}{(x^2+y^2+z^2)^{5/2}} \tag{53}$$

$$\nabla\cdot\mathbf{E} = \frac{q}{4\pi\varepsilon_0}\frac{2(x^2+y^2+z^2)-2(x^2+y^2+z^2)}{(x^2+y^2+z^2)^{5/2}} \tag{54}$$

$$\boxed{\nabla\cdot\mathbf{E} = 0, \quad \text{for } r \neq 0} \tag{55}$$

Thus, the divergence is zero everywhere (except at the charge location).

Since, the net charge density is zero everywhere so by Gauss's Law also the divergence in zero.

6.5 Verify whether the curl of the electric field is zero.

The curl of a vector field $\mathbf{E} = (E_x, E_y, E_z)$ in Cartesian coordinates is given by:

$$\nabla\times\mathbf{E} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix} \tag{56}$$

Expanding this determinant:

$$\nabla\times\mathbf{E} = \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z}\right)\hat{i} + \left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x}\right)\hat{j} + \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y}\right)\hat{k} \tag{57}$$

The electric field due to a point charge q located at the origin is:

$$\mathbf{E} = \frac{q}{4\pi\varepsilon_0}\frac{\mathbf{r}}{r^3} \tag{58}$$

Expressing components:

$$E_x = \frac{q}{4\pi\varepsilon_0}\frac{x}{r^3}, \quad E_y = \frac{q}{4\pi\varepsilon_0}\frac{y}{r^3}, \quad E_z = \frac{q}{4\pi\varepsilon_0}\frac{z}{r^3} \tag{59}$$

where $r = \sqrt{x^2 + y^2 + z^2}$.
Now, we compute the partial derivatives:

$$\frac{\partial E_z}{\partial y} = \frac{\partial}{\partial y}\left(\frac{q}{4\pi\varepsilon_0}\frac{z}{r^3}\right) \tag{60}$$

Using the chain rule:

$$\frac{\partial}{\partial y}\left(\frac{z}{r^3}\right) = z \cdot \frac{\partial}{\partial y}\left(r^{-3}\right) + r^{-3} \cdot \frac{\partial z}{\partial y} \tag{61}$$

Since $\frac{\partial z}{\partial y} = 0$, only the first term remains.
Similarly, for $\frac{\partial E_y}{\partial z}$:

$$\frac{\partial E_y}{\partial z} = \frac{\partial}{\partial z}\left(\frac{q}{4\pi\varepsilon_0}\frac{y}{r^3}\right) \tag{62}$$

Using the same method:

$$\frac{\partial}{\partial z}\left(\frac{y}{r^3}\right) = y \cdot \frac{\partial}{\partial z}\left(r^{-3}\right) + r^{-3} \cdot \frac{\partial y}{\partial z} \tag{63}$$

Since $\frac{\partial y}{\partial z} = 0$, only the first term remains.
Comparing both terms:

$$\frac{\partial E_z}{\partial y} = \frac{\partial E_y}{\partial z} \tag{64}$$

Thus, the first component of $\nabla \times \mathbf{E}$ is zero.
Repeating similar calculations for the remaining components, we find:

$$\frac{\partial E_x}{\partial z} = \frac{\partial E_z}{\partial x}, \quad \frac{\partial E_y}{\partial x} = \frac{\partial E_x}{\partial y} \tag{65}$$

Since all terms satisfy:

$$\left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z}\right) = \left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x}\right) = \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y}\right) = 0 \tag{66}$$

we conclude that:

$$\boxed{\nabla \times \mathbf{E} = \mathbf{0}} \tag{67}$$

7 Given the spherically symmetric potential function in free space is: $V(r) = V_0 e^{-r/a}$ where V_0 and a are constants. We determine:

7.1 The electric field E at $r = a$

The electric field is obtained using:

$$\mathbf{E} = -\nabla V \tag{68}$$

In spherical coordinates, the gradient of a radially symmetric potential is:

$$E_r = -\frac{dV}{dr} \tag{69}$$

Taking the derivative:

$$\frac{dV}{dr} = -\frac{V_0}{a} e^{-r/a} \tag{70}$$

Thus, the radial electric field is:

$$\boxed{E_r = \frac{V_0}{a} e^{-r/a}} \tag{71}$$

At $r = a$:

$$\boxed{E_r(a) = \frac{V_0}{a} e^{-1}} \tag{72}$$

7.2 The charge density ρ_v at $r = a$

Using Gauss's Law in differential form:

$$\nabla \cdot \mathbf{E} = \frac{\rho_v}{\varepsilon_0} \tag{73}$$

For a spherically symmetric field:

$$\nabla \cdot \mathbf{E} = \frac{1}{r^2} \frac{d}{dr} \left(r^2 E_r \right) \tag{74}$$

Substituting E_r:

$$\frac{d}{dr} \left(r^2 E_r \right) = \frac{d}{dr} \left(r^2 \frac{V_0}{a} e^{-r/a} \right) \tag{75}$$

$$= \frac{V_0}{a} \frac{d}{dr} \left(r^2 e^{-r/a} \right) \tag{76}$$

Using the product rule:

$$\frac{d}{dr}(r^2 e^{-r/a}) = 2re^{-r/a} - \frac{r^2}{a}e^{-r/a} \tag{77}$$

$$= e^{-r/a}\left(2r - \frac{r^2}{a}\right) \tag{78}$$

Thus:

$$\nabla \cdot \mathbf{E} = \frac{1}{r^2}\frac{V_0}{a}e^{-r/a}(2r - r^2/a) \tag{79}$$

Using Gauss's law:

$$\boxed{\rho_v = \varepsilon_0 \frac{V_0}{a}e^{-r/a}\frac{(2r - r^2/a)}{r^2}} \tag{80}$$

At $r = a$:

$$\rho_v(a) = \varepsilon_0 \frac{V_0}{a}e^{-1}\frac{(2a - a)}{a^2} \tag{81}$$

$$\rho_v(a) = \varepsilon_0 \frac{V_0}{a^3}e^{-1}a \tag{82}$$

$$\boxed{\rho_v(a) = \varepsilon_0 \frac{V_0}{a^2}e^{-1}} \tag{83}$$

7.3 The total charge enclosed.

Using Gauss's Law in integral form:

$$\oint_S \mathbf{E} \cdot d\mathbf{S} = \frac{Q}{\varepsilon_0} \tag{84}$$

For a sphere of radius r:

$$\oint_S E_r dS = E_r(r) \cdot 4\pi r^2 \tag{85}$$

$$4\pi r^2 E_r = \frac{Q}{\varepsilon_0} \tag{86}$$

Substituting $E_r = \frac{V_0}{a}e^{-r/a}$:

$$4\pi r^2 \frac{V_0}{a}e^{-r/a} = \frac{Q}{\varepsilon_0} \tag{87}$$

Solving for Q:

$$\boxed{Q = 4\pi\varepsilon_0 r^2 \frac{V_0}{a}e^{-r/a}} \tag{88}$$

Substituting $r = a$:

$$Q = 4\pi\varepsilon_0 a^2 \frac{V_0}{a}e^{-1} \tag{89}$$

$$\boxed{Q = 4\pi\varepsilon_0 V_0 a e^{-1}} \tag{90}$$

8 A parallel-plate capacitor has plates located at $z = 0$ and $z = d$. The region between the plates is filled with a material that contains a uniform volume charge density ρ_0 C/m^3 and has permittivity ϵ. Both plates are held at ground potential.

8.1 Determine the electric field intensity E between the plates.

To find the electric field, we use Gauss's Law:

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{enc}}}{\varepsilon} \tag{91}$$

Consider a Gaussian surface as a pillbox of area A extending from $z = 0$ to some height z. The charge enclosed in this volume is:

$$Q_{\text{enc}} = \rho_0 A z \tag{92}$$

By symmetry, $\mathbf{E}$ is uniform in the z-direction:

$$EA = \frac{\rho_0 A z}{\varepsilon} \tag{93}$$

Canceling A:

$$\boxed{E(z) = \frac{\rho_0}{\varepsilon} z} \tag{94}$$

8.2 Determine the potential field between the plates.

The electric field is given by:

$$E(z) = -\frac{dV}{dz} \tag{95}$$

Differentiating $V(z)$:

$$E(z) = -\left(\frac{\rho_0}{2\varepsilon} d - \frac{\rho_0}{\varepsilon} z\right) \tag{96}$$

$$\boxed{E(z) = \frac{\rho_0}{\varepsilon}\left(z - \frac{d}{2}\right)} \tag{97}$$

8.3 Repeat parts (a) and (b) for the case where the plate at z = d is raised to a potential V0, with the plate at z = 0 grounded.

Now, the boundary conditions are:
$V(0) = 0$ and $V(d) = V_0$

8.3.1 Potential calculation

We follow the same steps as before, using the general form:

$$V(z) = -\frac{\rho_0}{2\varepsilon}z^2 + C_1 z + C_2 \tag{98}$$

From $V(0) = 0$, we again get $C_2 = 0$, so:

$$V(z) = -\frac{\rho_0}{2\varepsilon}z^2 + C_1 z \tag{99}$$

Using $V(d) = V_0$:

$$V_0 = -\frac{\rho_0}{2\varepsilon}d^2 + C_1 d \tag{100}$$

Solving for C_1:

$$C_1 = \frac{\rho_0}{2\varepsilon}d + \frac{V_0}{d} \tag{101}$$

Thus, the potential is:

$$\boxed{V(z) = \left(\frac{\rho_0}{2\varepsilon}d + \frac{V_0}{d}\right) z - \frac{\rho_0}{2\varepsilon}z^2} \tag{102}$$

8.3.2 Electric Field for $V(d) = V_0$

Again, differentiating:

$$E(z) = -\frac{dV}{dz} \tag{103}$$

$$E(z) = -\left(\frac{\rho_0}{2\varepsilon}d + \frac{V_0}{d} - \frac{\rho_0}{\varepsilon}z\right) \tag{104}$$

$$\boxed{E(z) = \frac{\rho_0}{\varepsilon}\left(z - \frac{d}{2}\right) - \frac{V_0}{d}} \tag{105}$$